



D Y Patil College of Engineering, Akurdi-Pune

Department of Robotics and Automation

FAB Lab (Basic Robotics Workshop)

Project Proposal

Title of Project : EviLog - Evidence Tracking System.

Abstract :

EviLog is a low-cost smart evidence monitoring system developed to improve the identification, tracking, and accountability of physical evidence handled within police stations, forensic laboratories, and other authorized locations. In conventional evidence management, information regarding the storage, transfer, handling, and return of evidence may depend heavily on manual registers and paperwork. Maintaining these records accurately becomes challenging when evidence is transferred between multiple personnel and locations. EviLog proposes an embedded system that creates a digital and traceable record of these activities.

The system assigns a unique identification tag to each evidence package using **Radio Frequency Identification (RFID)** technology. Whenever an evidence item is moved, the RFID tag is scanned and the system records important information such as the evidence ID, handler, location, time of movement, type of transaction, and current status. An authorized user can therefore verify the latest location and custody status of an evidence item and review its previous movement history.

To improve monitoring, EviLog can incorporate additional features such as **tamper detection, unauthorized-movement alerts, and condition/status recording**. For example, if an evidence package is opened or moved unexpectedly, the system can generate an alert for further verification. A local storage mechanism can maintain the transaction history even without continuous internet connectivity, making the prototype suitable for environments where network availability may be limited.

The proposed prototype can be implemented using affordable components such as an **ESP32 microcontroller, RFID reader, RFID tags, OLED/LCD display, buzzer, status LEDs, RTC module, and SD card storage**. The system demonstrates how embedded systems and RFID technology can be applied to improve evidence traceability without replacing established legal procedures for evidence handling.

Overall, EviLog aims to provide a **cost-effective, transparent, and easily demonstrable solution for digital evidence movement monitoring**, reducing dependence on manual tracking and helping authorized personnel maintain a reliable record of the chain of custody.

Team :

Sr. No.	Roll No.	Name	Signature
1.		Bhavik Bokde	
2.		Om Patil	
3.		Saee Patil	
4.		Tanuja Jadhav	
5.		Vedant Adsul	

Ms. D. V. Kotkar

Dr. N. K. Kamble

Project Guide

FAB Lab Co-ordinator

HOD (R&A)